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Data Flow Design

Overview

The data flow describes how data moves through different components of the Flood Prediction System.

Data Flow Steps

Step 1: Data Collection

Historical flood datasets are collected from reliable sources. The dataset contains environmental and geographical information related to previous flood events.

Step 2: Data Preprocessing

The collected data is cleaned and prepared for machine learning.

Operations include:

- Removing unwanted data
- Handling missing values
- Converting data formats
- Selecting important features

Step 3: Model Training

The processed dataset is divided into training and testing data.

Training data is used to train machine learning algorithms, while testing data is used to evaluate performance.

Step 4: Prediction

New user input data is provided to the trained model.

The model analyzes the input and predicts the flood risk.

Step 5: Result Display

The prediction result is displayed through the Flask web application.

Text Document

Data Flow Diagram

Dataset

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Data Preprocessing

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Feature Extraction

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ML Model Training

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Trained Model

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User Input

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Prediction

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Flood Risk Result